Intelligent Electromagnetic Compatibility Diagnosis and Management With Collective Knowledge Graphs and Machine Learning

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***Abstract*—The explosive growth of electronic devices brings a soaring demand for rapid electromagnetic compatibility (EMC) diagnosis. However, there is a significant learning curve for the electrical engineers to apply EMC knowledge. In this article, an efficient EMC diagnosis and management methodology was proposed, which provided a fast way for EMC analysis in seconds other than traditional simulation or calculation. The approach organized the EMC knowledges as knowledge graph composed by the interference/sensitive units, and mathematical set rules. The optimized graph structure is in form of rule, maxterm, basic unit, and entity layers. Based on the condensed relationships, it achieved high searching efficiency and graph expansibility. To facilitate the information retrieval from the knowledge graph, the interference/sensitive units and related parameters were acquired from interactive sessions, in which long short-term memory method was used to extract entities. The EMC specialized corpora were fed in training to enhance the accuracy of inference. Finally, the EMC diagnosis and management reports were automatically generated by knowledge graph searching application. The proposed method improved the calculation efficiency by three times. The storage of relationships and attributes of nodes was reduced by 76% and 60.7%. The identification accuracy was enhanced from 77.7% to 99.5%. The presented method is practically useful for EMC design in crosstalk analysis, radiated, and conducted interference diagnoses.**

***Index Terms*—Collective knowledge graph, electromagnetic compatibility (EMC) management, interactive sessions, long short-term memory (LSTM) method, mathematical sets.**

I. Introduction

With the application of advanced chips, the electronic products tend to be compact and highly integrated. Many components and traces are integrated in a small space, which may cause severe electromagnetic compatibility (EMC) problems. According to published literatures [1]– [5], EMC diagnosis and management are mainly based on analytical method, hybrid method of circuit parameter extraction and full wave simulation, and measurement. However, analytical results are often obtained under some assumptions and cannot deal with complex structures [1]. For the electronic devices with multilayer printed circuit board (PCB), which has many components, chips, and traces, hybrid method of circuit parameter extraction and full wave simulation faces the difficulties in modeling, parameter extraction, and calculation. Even plenty of accelerated algorithms such as time-domain surface integral technique [2] and time-domain layered finite-element reduction-recovery method [3] are applied, the computational complexity is still enormous, which causes only partial or key component simulations in PCB are feasible. Moreover, measurement is helpful for interference diagnosis but it is expensive and cannot be applied in advance before the device is produced [4], [5]. Thus, intelligent and fast EMC diagnosis and management have attracted a lot of concerns [6]–[15]. Knowledge-based systems were firstly proposed for EMC analysis in 1959 [6]. In 1990s, knowledge base was widely used in EMC diagnosis [7]–[11]. Knowledge graph was recommended as a tool for industry, engineers, and researchers to share EMC experience [7]. Surekha *et al.* [8] proposed the frame-based knowledge system to express the EMC knowledge. Choo *et al.* [9] established the knowledge base with some simple EMC rules for EMC analysis. After that, however, the application for knowledge base in EMC diagnosis and management faced some problems owing to limitation of the machine learning and question and answering (QA) technologies. Many EMC diagnosis and management researches turned to artificial neural network (ANN) and support vector machines (SVMs) methods [10]–[15]. Devabhaktuni *et al.* [10] presented a neural network modeling approach for the electromagnetic interference analysis of PCB and shielding enclosures. McCormick *et al.* [11] applied ANN to predict the radiation from cables. Radial basis function neural network was proposed for the behavioral modeling of nonlinear circuit elements [12]. Chahine *et al.* [13] utilized neural networks for predicting the integrated circuit susceptibility to conducted electromagnetic disturbances. Echo state network was utilized to model the integrated circuit signals in the time domain for the direct power injection immunity test [14]. SVMs were used to model electromagnetic immunity of integrated circuit [15]. Recently, more and more machine learning models

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have been applied for EMC diagnosis and management with the contribution of high-performance graphic processing unit (GPU) [16], [17]. Meanwhile, the leaps-and-bounds of machine learning and QA technologies have significantly promoted the application of knowledge graph in different industries [18], including in medical industry, urban management, privacy protection, and EMC field [19]–[23].

An intelligent EMC management method with fully connected knowledge graph was first proposed in [23]. The interference and sensitive units in PCB were analyzed and stored as entities in the knowledge graph, and triple functions were defined to associate these entities with interference relationships. In the fully connected knowledge graph, all interference and sensitive units are connected, and EMC diagnosis needs several rounds of interactive sessions to determine whether there is interference relationship between two units. Although this fully connected knowledge graph owns simple structure, it has four major defects. First, there are abundant redundant relationships between nodes in the graph, which consume unnecessary searching time and storage space. Second, it needs several rounds of interactive sessions to acquire all the parameters in a relationship function [23]. Furthermore, it should traverse all the graph to find a node or relationship, leading to low efficiency. Finally, expansibility of fully connected knowledge graph is poor since adding or deleting a node or relationship requires significant change.

In this article, collective knowledge graph was established, and accurate entity extraction model was developed for EMC diagnosis. Mathematical set rules were first proposed to establish the knowledge graph for simplification and enhance searching efficiency. Moreover, improved machine learning model and EMC specialized corpora were developed to enhance the accuracy. The proposed method is applicable for EMC diagnosis and management with high efficiency and accuracy. Two typical EMC cases as crosstalk and radiation interference analyses are given.

The rest of this article is structured as follows. Section II briefly introduces the basic principle of proposed method. Section III presents the knowledge graph constructed with mathematical sets. Section IV elaborates improved entity extraction method by using machine learning. Section V presents the experiment and results. Finally, Section VI concludes this article.

II. Basic Principle

The intelligent EMC diagnosis and management system is composed mainly by two parts, i.e., knowledge graph and QA, which are used for storing EMC knowledges and acquiring the information of specific EMC problems, shown as the upper and bottom parts in Fig. 1, respectively.

First and foremost, the interference and sensitive units and related EMC rules are determined based on design documents, engineer's experience, and EMC guidelines through EMC analysis. Next, these units are stored as entities, which are expressed as nodes in the knowledge graph based on mathematical set and EMC rules. The relationships such as interference and affiliation between nodes are expressed with edges in the graph. Different

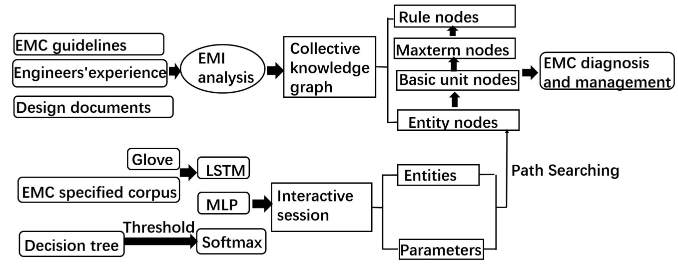

Fig. 1. Overall flowchart of the proposed method.

from traditional knowledge graph, which stores entities as individual nodes, the nodes in the collective knowledge graph in this article are grouped into entity, basic unit, maxterm and rule nodes by mathematical set operation, and affiliation relationship between nodes in different layers may exist.

The EMC diagnosis is the searching process in the knowledge graph. In other words, path searching is executed to determine whether there is a reachable path named as edge representing interference relationship between two entities.

A good knowledge graph has high searching efficiency and excellent data storage. Collective knowledge graph stores data according to set rules, which enhances the path searching efficiency in the EMC diagnosis.

Meanwhile, questions for EMC diagnosis and related information are inputted by engineers. Therefore, QA is important for system performance. In the QA system, the machine learning methods are arranged as a pipeline for entity extraction. The global vectors for word representation (GloVe) [20] is used to model the EMC specified corpora as vectors to feed the long short-term memory (LSTM) method [21], which generates the features of sentences as input of multilayer perceptron (MLP) and Softmax methods.

III. Collective Knowledge Graph

In the model proposed in [23], all the interference and sensitive modules and traces are stored as nodes, and two nodes are connected if there is interference relationship between them. This fully connected knowledge graph needs n rounds of interactive sessions to make decisions (n is the number of attributes of an entity). In this article, the collective knowledge graph is presented, which can reduce the rounds of interactive sessions to get results fast.

A. Interference and Sensitive Units

According to design documents and engineer's experience, major interference, and sensitive units of electronic devices can be determined. For simplification, commonly used electronic device as cell phones are taken as example and their interference and sensitive units are shown in Table I. Two levels are defined according to interference degree and sensitivity. The first level is assigned for the most interference or sensitive units. Interference modules include central processing unit (CPU), power management IC (PMIC), memory, camera, LCD, smart power amplifier (PA), and as well as Bluetooth, WiFi, and cellular

TABLE I
INTERFERENCE AND SENSITIVE UNITS

Classification	Entity
First level interference module	CPU, memory, PMIC, smart PA
Second level interference module	LCD, camera, Bluetooth transmitter, WIFI transmitter, cellular transmitter
First level sensitive module	GPS, FM, Bluetooth receiver, WIFI receiver, cellular receiver, Bluetooth antenna, WIFI antenna, cellular antenna, GPS antenna, FM antenna
Second level sensitive module	Microphone
First level interference trace	PMIC-CPU connection trace, PMIC-memory connection trace, PMIC-camera connection trace, PMIC-smart PA connection trace, PMIC-LCD connection trace, CPU-memory connection trace, CPU-camera connection trace, CPU-LCD connection trace
Second level interference trace	PMIC-battery connection trace, PMIC-speaker connection trace, PMIC-Bluetooth transmitter connection trace, PMIC-WIFI transmitter connection trace, PMIC-cellular transmitter connection trace
First level sensitive trace	CPU-Bluetooth clock trace, CPU-WIFI clock trace, CPU-cellular clock trace, CPU-FM clock trace, CPU-GPS clock trace, CPU-microphone clock trace, crystal-Bluetooth clock trace, crystal-WIFI clock trace, crystal-cellular clock trace, crystal-FM clock trace, crystal-GPS clock trace
Second level sensitive trace	PMIC-GPS connection trace, PMIC-Bluetooth receiver connection trace, PMIC-WIFI receiver connection trace, PMIC-cellular receiver connection trace, CPU-memory clock trace

TABLE II
LOGIC SYMBOLS OF ENTITIES

Classification	Attribute	Logic symbol
Interference or sensitive unit	Interference unit	$A(x)$
Interference or sensitive unit	Sensitive unit	$\overline{A(x)}$
Module or trace	Module	$B(x)$
Module or trace	Trace	$\overline{B(x)}$
Level	First level	$C(x)$
Level	Second level	$\overline{C(x)}$
Interference	Interference between x and y	$D(x, y)$
Interference	No interference between x and y	$\overline{D(x,y)}$
Ground	There is ground on the top of x	$E1(x)$
Ground	There is ground on the bottom of x	$E2(x)$
Ground	There is ground on the left of x	$E3(x)$
Ground	There is ground on the right of x	$E4(x)$
Ground	There is no ground around x	$\overline{E1(x) \vee E2(x) \vee E3(x) \vee E4(x)}$
Distance	The distance between x and y is smaller than d	$F(x, y, d)$
Distance	The distance between x and y is smaller than $3W$	$F(x, y, 3W)$
Distance	The distance between x and y is larger than $3W$	$\overline{F(x,y,3W)}$
Distance	The distance between x and y is smaller than $10W$	$F(x, y, 10W)$
Distance	The distance between x and y is larger than $10W$	$\overline{F(x,y,10W)}$
Distance	The distance between x and y is larger than $3W$ and smaller than $10W$	$\overline{F(x,y, 3W)} \wedge F(x,y, 10W)$
Distance	The distance between x and y is smaller than $\lambda/10$	$F(x,y,\lambda/10)$
Distance	The distance between x and y is larger than $\lambda/10$	$\overline{F(x,y,\lambda/10)}$

transmitters. Sensitive modules include radio frequency (RF) module receivers (e.g., Bluetooth, WiFi, and cellular receivers), global positioning system (GPS), frequency modulation (FM), microphone, and RF antennas (Bluetooth, WiFi, cellular, FM, and GPS). Moreover, interference traces include all the traces connecting the interference modules, such as connection line between PMIC and CPU. The sensitive traces consist of all the traces connected with the sensitive modules or crystal oscillator.

The interference between two units is determined by multiple factors, which can be obtained through EMC analysis. In this case, they are distance, operating frequency, and shielding condition. These factors can be taken as intersective and complement set operations in math. It means if there is no interference between two units, these two units should belong to two unrelated sets. By this method, multiple rounds of interactive sessions are changed to mathematical set calculation with less rounds in collective knowledge graph.

B. Nodes in the Collective Knowledge Graph

The collective knowledge graph is composed by four layers as rule, maxterm, basic unit, and entity layers. All the nodes in rule, maxterm, and basic unit layers are called collective nodes. The nodes in rule layer represent logical rules and related conclusions, and nodes in maxterm layer represent the maxterms of the simplest intersection or union of sets. Moreover, nodes in basic unit layer represent the indivisible units involved in maxterms, and nodes in entity layer represent the basic entity units, which correspond to interference and sensitive units for different EMC scenarios, referring to modules, and traces in PCB of cell phones in this article.

The collective nodes are obtained by converting the EMC rules to logic expressions, which are then simplified into simplest intersection or union of sets. When an entity is added into the knowledge graph, first, the affiliated relationship between the entity and basic unit layer node is established according to entity's attribute. Next, whether there is a path between the entity and each maxterm node is determined. If yes, the relationship between the entity and rule layer is established. The logic symbols of entities are shown in Table II.

In Table II, W is trace width,$\lambda$is the wavelength of sensitive unit, $\wedge$ is the intersection of sets, and $\vee$ is the union of sets.

The logical expressions are acquired based on EMC rules. For example, one EMC rule is that the interference module may interfere the sensitive module when the distance between them is smaller than $\lambda/10$ without shielding. According to Table II, the logic symbols of interference and sensitive modules are $A(x) \wedge B(x)$ and $\overline{A(y)} \wedge B(y)$, respectively. If the distance between these two modules is smaller than $\lambda/10$, the corresponding

TABLE III
LOGIC EXPRESSIONS OF SOME EMC RULES

Rule	Logic expressions
Type: Interference module Type: Sensitive module Distance: Smaller than $\lambda/10$ Conclusion: Interference	$(A(x) \wedge B(x)) \wedge (\overline{A(y)} \wedge B(y)) \wedge F(x,y,\lambda/10) \rightarrow D(x,y)$
Type: First level interference trace Condition: No ground on the top and bottom layers Type: First level sensitive trace Condition: No ground all around Distance: Smaller than 10 W Conclusion: Interference	$(A(x) \wedge \overline{B(x)} \wedge C(x) \wedge (\overline{E1(x)} \vee \overline{E2(x)})) \vee (\overline{A(y)} \wedge \overline{B(y)} \wedge C(y) \wedge (\overline{E1(y)} \vee \overline{E2(y)} \vee \overline{E3(y)} \vee \overline{E4(y)})) \vee F(x,y,10W) \rightarrow D(x,y)$
Type: First level interference trace Condition: No ground on the top and bottom layers Type: Second level sensitive trace Condition: No ground on the top and bottom layers Distance: Smaller than 10 W Conclusion: Interference	$(A(x) \wedge \overline{B(x)} \wedge C(x) \wedge (\overline{E1(x)} \vee \overline{E2(x)})) \vee (\overline{A(y)} \wedge \overline{B(y)} \wedge \overline{C(y)} \wedge (\overline{E1(y)} \vee \overline{E2(y)})) \vee F(x,y,10W) \rightarrow D(x,y)$
Type: Second level interference trace Condition: No ground on the top or bottom layers Type: First level sensitive trace Condition: No ground all around Distance: Smaller than 10W Conclusion: Interference	$(A(x) \wedge \overline{B(x)} \wedge \overline{C(x)} \wedge (\overline{E1(x)} \wedge \overline{E2(x)})) \vee (\overline{A(y)} \wedge \overline{B(y)} \wedge C(y) \wedge (\overline{E1(y)} \vee \overline{E2(y)} \vee \overline{E3(y)} \vee \overline{E4(y)})) \vee F(x,y,10W) \rightarrow D(x,y)$
Type: Second level interference trace Condition: No ground on the top or bottom layers Type: Second level sensitive trace Condition: No ground on the top and bottom layers Distance: Smaller than 10W Conclusion: Interference	$(A(x) \wedge \overline{B(x)} \wedge \overline{C(x)} \wedge (\overline{E1(x)} \wedge \overline{E2(x)})) \vee (\overline{A(y)} \wedge \overline{B(y)} \wedge \overline{C(y)} \wedge (\overline{E1(y)} \vee \overline{E2(y)})) \vee F(x,y,10W) \rightarrow D(x,y)$
Type: Second level sensitive trace Condition: No ground on the top and bottom layers Type: Second level sensitive trace Condition: No ground on the top and bottom layers Distance: Smaller than 3W Conclusion: Interference	$(\overline{A(x)} \wedge \overline{B(x)} \wedge \overline{C(x)} \wedge (\overline{E1(x)} \vee \overline{E2(x)})) \vee (\overline{A(y)} \wedge \overline{B(y)} \wedge \overline{C(y)} \wedge (\overline{E1(y)} \vee \overline{E2(y)})) \vee F(x,y,3W) \rightarrow D(x,y)$

logic symbol is $F(x, y, \lambda/10)$. Moreover, if there is interference between two modules, the logic symbol is $D(x, y)$. Therefore, the logic expression of this EMC rule is$(A(x) \wedge B(x)) \wedge (\overline{A(y)} \wedge B(y)) \wedge F(x, y, \lambda/10) \rightarrow D(x, y)$.

Following this method, the EMC rules and corresponding logic expressions are listed in Table III.

After the logic symbols and expressions of entities and EMC rules are obtained, the collective nodes of rule, maxterm, and basic unit layers are easy to be generated. For example, an EMC rule is that the first level interference trace may interfere the first level sensitive trace when the first level interference trace is not surrounded by ground on the top and bottom layers, or the first level sensitive trace is not surrounded by ground all around, or the distance between these two traces is smaller than 10W. In this rule, there are two entities as first level interference and sensitive traces. This rule is divided into three rules as follows.

1) The first level interference trace is not surrounded by ground on the top and bottom layers.
2) The first level sensitive trace is not surrounded by ground all around.
3) The distance between these two traces is smaller than 10 W.

According to Table II, the entity in rule 1) is expressed as

$$A(x) \wedge \overline{B(x)} \wedge C(x) \wedge \left(\overline{E1(x) \wedge E2(x)}\right)$$
$$= A(x) \wedge \overline{B(x)} \wedge C(x) \wedge \left(\overline{E1(x)} \vee \overline{E2(x)}\right). \quad (1)$$

The entity involved in rule 2) is expressed as

$$\overline{A(y)} \wedge \overline{B(y)} \wedge C(y)$$
$$\wedge \left(\overline{E1(y) \wedge E2(y) \wedge E3(y) \wedge E4(y)}\right)$$
$$= \overline{A(y)} \wedge \overline{B(y)}$$
$$\wedge C(y) \wedge \left(\overline{E1(y)} \vee \overline{E2(y)} \vee \overline{E3(y)} \vee \overline{E4(y)}\right). \quad (2)$$

The rule 3) is expressed as $F(x, y, 10\mathrm{W})$.

The first level interference trace may interfere the first level sensitive trace once one rule is true. The logic expression for this rule is

$$\left(A(x) \wedge \overline{B(x)} \wedge C(x) \wedge \left(\overline{E1(x)} \vee \overline{E2(x)}\right)\right)$$
$$\vee \left(\overline{A(y)}\right.$$
$$\left. \wedge \overline{B(y)} \wedge C(y) \wedge \left(\overline{E1(y)} \vee \overline{E2(y)} \vee \overline{E3(y)} \vee \overline{E4(y)}\right)\right)$$
$$\vee F(x, y, 10\mathrm{W}) \rightarrow D(x, y). \quad (3)$$

According to (3), the nodes in rule layer include$A(x) \wedge \overline{B(x)} \wedge C(x)$, $\overline{A(y)} \wedge \overline{B(y)} \wedge C(x)$, $\overline{A(y)} \wedge \overline{B(y)} \wedge C(x) \wedge (\overline{E1(y)} \vee \overline{E2(y)} \vee \overline{E3(y)} \vee \overline{E4(y)})$,$D(x, y)$, and$A(x) \wedge \overline{B(x)} \wedge C(x) \wedge (\overline{E1(x)} \vee \overline{E2(x)})$. The nodes in maxterm layer include$\overline{A(y)} \wedge \overline{B(y)} \wedge C(x)$, $A(x) \wedge \overline{B(x)} \wedge C(x)$, $\overline{E1(x)} \vee \overline{E2(x)}$, and $\overline{E1(y)} \vee \overline{E2(y)} \vee \overline{E3(y)} \vee \overline{E4(y)}$. The nodes in basic unit layer are composed by $A(x)$, $\overline{A(x)}$, $B(x)$, $\overline{B(x)}$, $C(x)$, $\overline{C(x)}$, $E1(x)$, $\overline{E1(x)}$, $E2(x)$, $\overline{E2(x)}$, $E3(x)$, $\overline{E3(x)}$, $E4(x)$, $\overline{E4(x)}$.

Following this method, a collective knowledge graph is generated with 33 collective nodes, 10 nodes in rule layer, 9 nodes in maxterm layer, and 14 nodes in basic unit layer.

C. Relationships Between Nodes

Relationships in the graph include two types as relationships between collective nodes and that between entity and collective nodes.

TABLE IV
DISTANCE FUNCTION RELATIONSHIPS

Entity x	Layer	Entity y	Layer	Distance function relationships
Interference module	Rule layer	Sensitive module	Rule layer	Function (x, y, $\lambda/10$)
First level interference trace	Rule layer	First level sensitive trace	Rule layer	Function (x, y, $10W$)
First level interference trace	Rule layer	Second level sensitive trace	Rule layer	Function (x, y, $10W$)
Second level interference trace	Rule layer	First level sensitive trace	Rule layer	Function (x, y, $10W$)
Second level interference trace	Rule layer	Second level sensitive trace	Rule layer	Function (x, y, $10W$)
Second level sensitive trace	Rule layer	Second level sensitive trace	Rule layer	Function (x, y, $3W$)

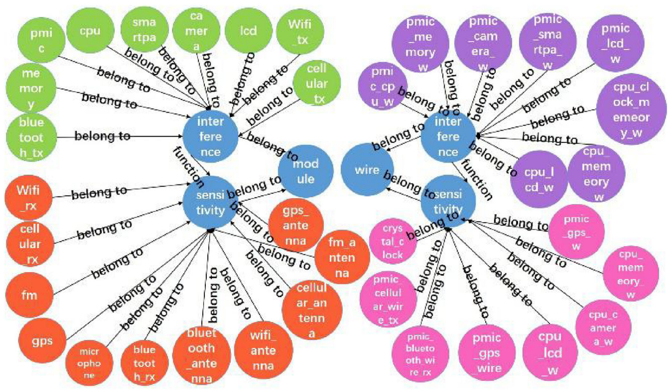

Fig. 2. Collective knowledge graph.

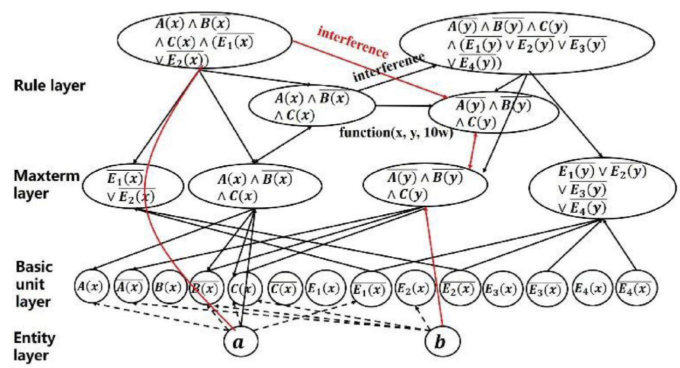

Fig. 3. EMC diagnosis by using the collective knowledge graph.

The relationships between collective nodes are composed by affiliation, distance function, and interference relationships.

1) Affiliation relationship. The rule nodes belong to corresponding maxterm nodes, and maxterm nodes belong to corresponding basic unit nodes.
2) Distance function relationship. Distance is the only input parameter of the distance function relationship. However, there are several input parameters for a function in the fully connected knowledge graph, such as distance, ground, and shielding condition. Compared with fully connected knowledge graph, collective knowledge graph achieved dimension reduction by saving the multiple parameter information in the collective nodes. Table IV presents the entity locations and the relationships between entities.
3) Interference relationship. The interference relationship refers to electromagnetic interference between two nodes. For example, if an entity belongs to the set which is the first level interference trace not surrounded by ground on the top and bottom layers, and another entity belongs to the first level sensitive trace set, then there is interference between these two entities. This means, the nodes belonging to these two sets have interference relationship.

Moreover, the relationships between entity nodes and the collective nodes are determined based on the attributes of entity nodes. For example, entity nodes CPU, GPS, and PMIC belong to the module of basic unit nodes.

D. Collective Knowledge Graph and Application

The collective knowledge graph is generated by the design documents, EMC and mathematical set rules, which includes 87 nodes, 118 relationship functions, and 1183 node attributes, part as shown in Fig. 2.

The intelligent EMC diagnosis and management is conducted by the collective knowledge graph presented in Fig. 2. For instance, if we want to determine whether EMI exists between "PMIC-LCD connection trace" and "CPU-Bluetooth clock trace", first we find the entity "PMIC-LCD connection trace" belongs to first level interference trace, and the entity "CPU-Bluetooth clock trace" belongs to first level sensitive trace. Then, the system would determine whether there is a distance function between these two entities. If no, there is no interference between two entities. In the graph, there is a distance Function (x, y, 10W) between them. If the distance between two entities is smaller than 10W, there is interference. On the other hand, if the distance is larger than 10W, the grounding condition should be determined. If there is no ground on the top of "PMIC-LCD connection trace", then the entity "PMIC-LCD connection trace" belongs to $\overline{E1(x)}$. In the graph, entity "PMIC-LCD connection trace" can reach the maxterm nodes $A(x) \wedge \overline{B(x)} \wedge C(x)$ and $\overline{E1(x)} \vee \overline{E2(x)}$ simultaneously. Thus, entity "PMIC-CPU connection trace" belongs to the set $A(x) \wedge \overline{B(x)} \wedge C(x) \wedge\ (\overline{E1(x)} \vee \overline{E2(x)})$, which means entity node "PMIC-LCD connection trace" belongs to rule node "$A(x) \wedge \overline{B(x)} \wedge C(x) \wedge\ (\overline{E1(x)} \vee \overline{E2(x)})$". Therefore, there is a path presenting interference relationship from entity node "PMIC-LCD connection trace" to entity node "CPU-Bluetooth clock trace", which is shown as the red path in Fig. 3, so we can infer that there is interference between "PMIC-LCD connection trace" and "CPU-Bluetooth clock trace".

This example shows the process of EMC diagnosis and management by searching the entities and their relationships in the collective knowledge graph. However, the investigated entities and their information should be extracted through QA, which is illustrated in Section IV.

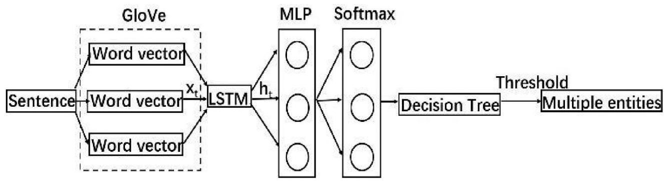

Fig. 4. Flowchart of the QA steps.

IV. Machine Learning in QA System

During the EMC diagnosis, the specific EMC problem is inputted by engineer in the QA system where the machine learning is applied to extract the entities accurately. The steps are as follows. First, the natural language is mapped to vectors by GloVe [24]. Next, features of sentences are extracted by using LSTM method [25]. Finally, the entities are classified and extracted by using MLP and Softmax methods. These steps are illustrated in Fig. 4.

A. Mapping the Natural Language to Vectors

Mapping the natural language to vectors is the coding process. The words are expressed as fixed length continuous dense vectors. Since high speed is required in the interactive sessions, the distributed representation method based on language models is applied for coding, which utilizes the context information for prediction. In this article, GloVe model, which is a word representation tool based on count-based and overall statistics, is used to generate word vectors. It maps words to vectors, which capture some semantic characteristics between words, such as similarity and analogy. There are mainly four steps to generate the word vectors.

1) Constructing Co-Occurrence Matrix: The co-occurrence matrix is constructed based on corpora, and the element X_{ij} in the matrix represents the co-occurrence times for word *i* and its context word *j* in a context window.

2) Constructing the Relationships Between Word Vectors and Co-Occurrence Matrix: Based on (4), the relationships between word vectors and co-occurrence matrix are constructed

$$w_i^T w_j + b_i + b_j = \log(X_{ij}) \tag{4}$$

where w_i^T and w_j are the word vectors needed to be solved, and b_i and b_j are the biases of the word vectors.

3) Generating the Loss Function:

$$J = \sum_{i,j=1}^{V} f(X_{ij})\left(w_i^T w_j + b_i + b_j - \log(X_{ij})\right)^2. \tag{5}$$

The basic form of this loss function is mean square loss function. The weight function $f(x_{ij})$ is given as follows:

$$f(x_{ij}) = \begin{cases} (x_{ij}/x_{\max})^a & x_{ij} < x_{\max} \\ 1 & \text{otherwise} \end{cases} \tag{6}$$

where a and $x_{\max}$ are equal to 0.75 and 100 in this model.

4) Outputting the Word Vectors: Although training corpora for GloVe model are huge, some EMC specialized vocabularies are not included in the thesaurus. In this article, an EMC specialized thesaurus is generated and added. If the coded word is in the EMC specialized thesaurus, one-dimensional (1-D) feature marked with 1 is added. Otherwise, 1-D feature marked with 0 is added. Each word is mapped to a 301-D vector with the Glove model, and a sentence with n words is expressed with a n×301-D matrix.

B. Extracting Features of Sentences by Using LSTM

In the interactive sessions, after the questions and answers inputted by users are mapped to vector matrix with GloVe, the LSTM is applied for feature extraction. LSTM is composed by input, output, and forget gates. For the inputted information, the forget gate determines, which information should be forgotten by

$$f_t = \sigma(W_f \bullet [C_{t-1}, x_t] + b_f) \tag{7}$$

where f_t, x_t, C_{t-1}, σ, W_f and b_f are activation vector, input word vector, cell state vector, sigmoid function, weight, and bias.

Then, the candidate vectors are generated based on the memory and input as follows:

$$i_t = \sigma(W_i \cdot [C_{t-1}, x_t] + b_i) \tag{8}$$

$$\tilde{C}_t = \tanh(W_c \cdot [C_{t-1}, x_t] + b_c) \tag{9}$$

where i_t is the activation vector of input gate, is the candidate vector, W_i and W_c are weights and b_i and b_c are biases.

Next, the new cell state C_t is added by forgetting the old knowledge and adding new knowledge by using

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t\,. \tag{10}$$

Finally, the output is obtained by

$$o_t = \sigma(W_o \cdot [C_{t-1}, x_t] + b_o) \tag{11}$$

$$h_t = o_t \times \tanh(C_t) \tag{12}$$

where W_o and b_o are weight and bias, o_t is the activation vector of output gate, and h_t is the output of the LSTM.

C. Entity Classification and Extraction by Using MLP and Softmax

After features are extracted by LSTM, MLP is applied for entity classification, and Softmax obtains the probability distribution of each entity, and outputs the entities larger than threshold. Since there may be several entities in a sentence, 0.5 threshold in binary classification is not applicable. In this article, decision tree is used for threshold selection. In the multiple classification, the sum of entity probability is equal to 1. Thus, a decision tree with depth 1 is applied, and the split condition of the tree is the threshold. By training with the decision tree, the optimized threshold is chosen as 0.172. Therefore, all the entities whose probability is larger than 0.172 are outputted.

D. Performance Analysis

The corpus is collected from design documentations of electronic products, such as files from Cadence software, including

TABLE V
TRAINING, TEST, AND EVALUATION DATASETS

Number of sentences	Training set	Test set	Evaluation set
12840	8988	1926	1926

TABLE VI
ACCURACY VERSUS NUMBER OF LSTM LAYERS

Number of LSTM layer	Accuracy	*P*	*R*	*F1*
1	75.3%	73.5%	73.5%	73.5%
2	76.1%	73.0%	74.1%	75.4%
3	70.9%	67.0%	67.6%	67.3%

12 840 sentences. As Table V shows, the total datasets are randomly divided into training, validation, and test sets, in which 70%, 15%, and 15% are for training, validation, and test, respectively.

In the interactive sessions, LSTM and MLP are important for entity extraction. Thus, the influences of the key parameters of the two methods on the performance are investigated. The performance is evaluated via accuracy, recall (*R*), precision (*P*), and *F1* measure.

F1 measure is to check the validity of our model, which combines *R* and *P* with an equal weight, i.e.,

$$F1 = \frac{2RP}{R+P} \tag{13}$$

where *P* is the number of correct results divided by the number of all returned results and *R* is the number of correct results divided by the number of results that should have been returned

$$P = \frac{TP}{TP+FP} \tag{14}$$

$$R = \frac{TP}{TP+FN} \tag{15}$$

where *TP* is the number of entities extracted correctly, *FP* is the number of entities identified incorrectly, and *FN* is the number of entities not extracted.

The influence of the number of LSTM layers is investigated. The number of LSTM layers varies from 1 to 3, and neurons in each layer are fixed to 200 for comparison. Meanwhile, three-layer MLP is applied, with 256 neurons in the hidden layer and the learning rate is 0.1.

The results in Table VI and Fig. 5(a)–(c) show that the two-layer LSTM achieves the best performance. The two-layer LSTM is better than one-layer LSTM because its feature exaction capability is stronger. However, the performance does not increase with the increase in layers because the performance of two-layer LSTM is better than that of three-layer LSTM. The reason is that the complexity of the model and parameters increases with the number of layers, and a complex model tends to be overfitting. The relationship between the numbers of parameters and layers is investigated by changing the number of LSTM layers. In this case, 599 889, 920 689, and 1 241 489 parameters exist in one-, two-, and three-layer LSTMs,

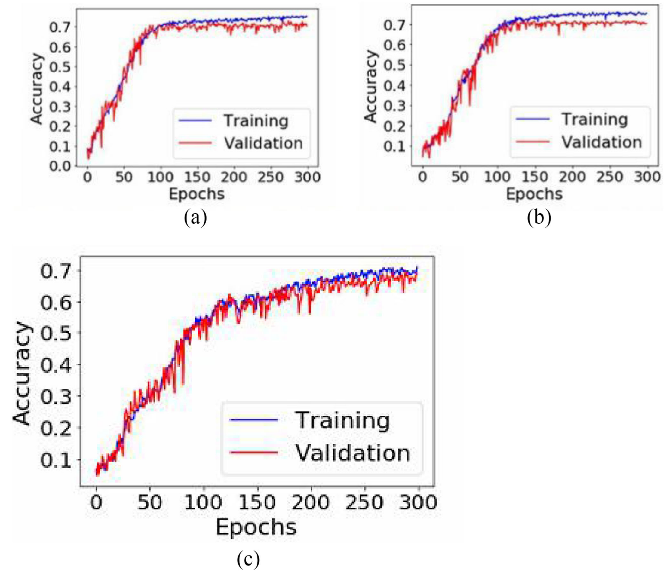

Fig. 5. (a) Accuracy versus epochs for one-layer LSTM. (b) Accuracy versus epochs for two-layer LSTM. (c) Accuracy versus epochs for three-layer LSTM.

TABLE VII
ACCURACY VERSUS HIDDEN SIZE OF LSTM

Hidden size	Accuracy	*P*	*R*	*F1*
50	62.2%	60.7%	61.3%	60.9%
100	56.5%	55.5%	56.0%	55.7%
200	76.4%	72.4%	72.6%	72.5%
400	79.4%	73.9%	74.1%	74.0%
800	75.9%	73.9%	74.1%	74.0%

respectively. If the training data are insufficient, realizing a similar distribution for the training and evaluation sets will be difficult. The increase in layers reduces the generalization capability of the model with numerous parameters. The model tends to be trapped in local optima, and the performance would deteriorate. For example, "WiFi transmitter" is predicted as "transmitter" in some cases by the three-layer LSTM model in the experiment. There is no rule of thumb on how many nodes (or hidden neurons) or how many layers one should choose. The design choice of making correct hyperparameter often a "block art that requires expert experiences" [26], [27]. In this article, we take a trial and error approach to get the best results for the dataset and problem.

As shown in Table VII and Fig. 6(a)–(e), the influence of the hidden size of LSTM on the performance is investigated. A two-layer LSTM is selected, and the hidden size of each layer varies from 50 to 800. The best performance is achieved when the hidden size is 400. The accuracy is low for the network with a small hidden size because the LSTM cannot extract all information of sentence with small feature dimensions. However, if the hidden size is excessively large, the training data are inadequate to train many parameters, thus decreasing the accuracy.

As shown in Table VIII and Fig. 7(a)–(d), the number of neurons of MLP varies from 64 to 512. The two-layer LSTM is

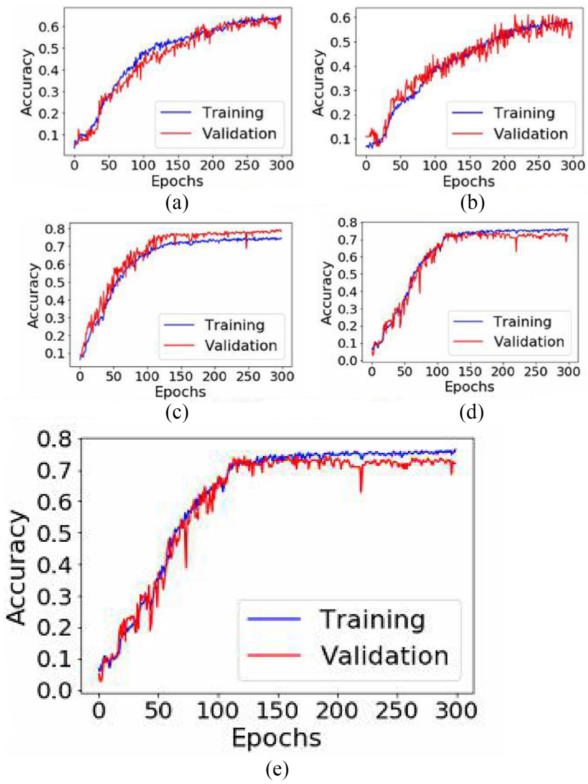

Fig. 6. (a) Accuracy versus epochs for 50 hidden size. (b) Accuracy versus epochs for 100 hidden size. (c) Accuracy versus epochs for 200 hidden size. (d) Accuracy versus epochs for 400 hidden size. (e) Accuracy versus epochs for 800 hidden size.

TABLE VIII
ACCURACY VERSUS NUMBER OF NEURONS IN THE HIDDEN LAYER

Number of neurons	Accuracy	P	R	F1
64	9.1%	7.9%	9.2%	8.5%
128	57.3%	55.7%	56.2%	56.0%
256	76.5%	73.3%	73.5%	73.4%
512	75.9%	74.0%	74.4%	74.2%

selected, and the hidden size of each layer is 400. The best performance is reached when the number of neurons is 256. The reason is that when the hidden size is smaller than 256, the excessive feature compression causes weak information processing ability. However, too many neurons may lead to be trapped in local minimum point and poor accuracy owing to insufficient training data. Moreover, as the running time increases with the number of neurons, 256 neurons in the hidden layer are optimal in this model in consideration of time consumption and accuracy.

As mentioned in part A, some EMC specialized vocabularies such as "crosstalk," "radiated emission," "conducted emission," and "shielding" are included in the thesaurus to enhance the accuracy. The performance of general and specialized corpora is compared. A two-layer LSTM with a hidden size of 400 is applied, and the neurons in the hidden layer of MLP are 256. Table IX shows that the accuracy of entity extraction, *P*, *R*, and *F1* measure is enhanced by 28%, 34%, 33.6%, and 33.7%, respectively, when some EMC specialized vocabularies are added.

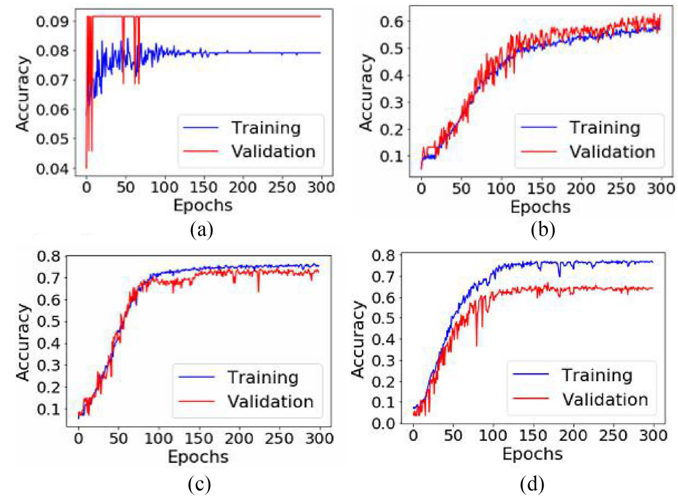

Fig. 7. (a) Accuracy versus epochs for 64 neurons. (b) Accuracy versus epochs for 128 neurons. (c) Accuracy versus epochs for 256 neurons. (d) Accuracy versus epochs for 512 neurons.

TABLE IX
PERFORMANCE COMPARISON BETWEEN GENERAL AND SPECIALIZED CORPORA

Type	Accuracy	*P*	*R*	*F1*
General corpus	77.7%	74.2%	74.4%	74.3%
Specialized corpus	99.5%	99.4%	99.4%	99.4%

V. RESULTS AND DISCUSSION

Two test cases were designed for EMC diagnosis and management to check the feasibility of proposed method, and calculation time, storage space, and accuracy were compared. These two cases are EMC problems frequently encountered in PCB design. One is the crosstalk analysis and the other is radiated interference diagnosis. The system was run in a workstation with NVIDIA GeForce GTX 1080 Ti and Intel Xeon(R) CPU E5-2603 v4 @ 1.70 GHz with 6 cores.

A. Crosstalk Between Two Parallel Traces

The test case is to determine the reason why a clock line connecting CPU and cellular in a PCB of cell phone is interfered, and there is no ground on the top and bottom layers of the line.

For a fully connected knowledge graph, the running process is as follows.

User questioning: Why is the clock line connecting cellular and CPU interfered?

System querying: What modules and wires are there around the CPU-cellular clock wire?

User answering: There is a line connecting PMIC and CPU.

System querying: What is the distance between the CPU-cellular clock wire and PMIC-CPU wire?

User answering: 20 mm.

System querying: What is the wire width?

User answering: 0.125 mm.

System querying: Is CPU-cellular clock wire encircled by ground layers?

User answering: No.

System querying: Are there any ground layers above and below the PMIC-CPU wire?

User answering: No.

System answering: The interference on CPU-cellular clock wire is caused by PMIC-CPU wire.

According to abovementioned process, fully connected knowledge graph needs six rounds of interactive sessions to make decisions. Nevertheless, there is less rounds of interactive sessions for collective knowledge graph shown as follows.

User questioning: Why is the clock line connecting cellular and CPU interfered?

System querying: What modules and wires are there around the CPU-cellular clock wire?

User answering: There is a line connecting PMIC and CPU.

System querying: What is the distance between the CPU-cellular clock wire and PMIC-CPU wire?

User answering: 20 mm.

System querying: What is the wire width?

User answering: 0.125 mm.

System querying: Is there any ground layer above the PMIC-CPU wire?

User answering: No.

System answering: The interference on CPU-cellular clock wire is caused by PMIC-CPU wire.

For the diagnosing process by using fully connected knowledge graph, first it determines the victim unit is the first level sensitive trace through user's question. Then, it is obtained that there is only a first level interference trace and no interference module around. Next, it is known that the distance between two traces meets 10W rule according to user's answer. Finally, according to the definition of F2(d, 10W, i1_ground, s1_ground), the system draws the conclusion there is interference between two traces [23]. Thus, 6 rounds of interactive sessions are required, and 136 575 s are spent.

For the collective knowledge graph, according to Fig. 3, PMIC-CPU wire belongs to entity a, and CPU-cellular clock wire belongs to entity b. There is no ground layer above entity a, so entity a belongs to $\overline{E1(x)}$. In the graph, entity a reaches maxterm nodes $A(x) \wedge \overline{B(x)} \wedge C(x)$ and $\overline{E1(x)} \vee \overline{E2(x)}$ simultaneously, which means entity node a belongs to rule node "$A(x) \wedge \overline{B(x)} \wedge C(x) \wedge (\overline{E1(x)} \vee \overline{E2(x)})$". Meanwhile, there is a path presenting interference relationship from entity node a to entity node b. Therefore, the conclusion "the interference on CPU-cellular clock wire is caused by PMIC-CPU wire" is obtained, and no other interactive sessions are needed. In fact, the interference diagnosing is the path searching in collective knowledge graph. It reduces the rounds of interactive sessions to enhance the efficiency. In total, 5 rounds of interactive sessions are required for the collective knowledge graph and 31 979 s are spent.

Table X compares time consumption and storage space between fully connected and collective knowledge graphs. The results show that the diagnosing time is reduced from 136 575 s to 31 979 s for the test case owing to the improvement of knowledge graph and machine learning model. For other applications, the fully connected knowledge graph needs all the parameters of the defined function to make the decision. If there are n parameters in the function, n rounds of interactive sessions are needed. However, the interactive sessions are less than n for collective knowledge graph. In worst case n rounds of interactive sessions are required, and in best case only one round of interactive session is needed. Therefore, collective knowledge graph shows higher efficiency comparing with fully connected knowledge graph. As for the storage space, the data size of collective knowledge graph is much smaller than fully connected knowledge graph. Since collective knowledge graph applies collective nodes, the number of nodes in collective knowledge graph increases slightly, but the relationships and node attributes are decreased by 76% and 60.7%, respectively.

TABLE X
COMPARISON BETWEEN PROPOSED METHOD AND METHOD IN [23]

Type	Time cost	Interactive rounds	Nodes	Relationships	Attributes
Fully connected knowledge graph	136576 s	6	54	491	3011
Collective knowledge graph	31979 s	5	87	118	1183

B. Radiated Interference Diagnosis Between Two Components

In the PCB, many components are accommodated in a small space. Therefore, some radiated components without careful EMC design may interfere with other sensitive components. In this part, we presented a case for the radiated interference analysis with the proposed method.

The diagnosis process with fully connected knowledge graph is as follows.

User questioning: Why is the GPS interfered?

System querying: What modules and wires are there around the GPS?

User answering: There is CPU nearby.

System querying: What is the distance between the CPU and GPS?

*User answering:*6 mm.

System querying: What frequency is the GPS?

User answering: The frequency is 1.5 GHz.

System querying: Is there a shield around the CPU?

User answering: No.

System querying: Is there a shield around the GPS?

User answering: No.

System answering: There is interference between CPU and GPS.

In the interference diagnosis process by using fully connected knowledge graph, the sensitive entity "GPS" is extracted from user's question with the method introduced in Section IV by using the combination method of Glove, LSTM, MLP, and Softmax. Then, the question about nearby modules and traces is asked based on templates. According to user's answer, the interference entity "CPU" is extracted by using the machine learning method. Subsequently, the function between CPU and GPS is found out in the knowledge graph, which is interference relationship $F(x, y, \lambda/10)$ in the knowledge graph. The questions about frequency of sensitive module and distance between CPU and GPS are asked based on templates to determine the parameters in $F(x, y, \lambda/10)$. According to the answers, the distance is smaller than 0.1λ. Therefore, interference may exist between CPU and GPS when there are no shields around them. Based on abovementioned process, six rounds of interactive sessions are required.

For the collective knowledge graph, the interactive sessions are as follows.

User questioning: Why is the GPS interfered?

System querying: What modules and wires are there around the GPS?

User answering: There is CPU nearby.

System querying: What is the distance between the CPU and GPS?

User answering: 6 mm.

System querying: What frequency is the GPS?

User answering: The frequency is 1.5 GHz.

System answering: There is interference between CPU and GPS.

The results show that the interactive sessions are reduced to four rounds from six rounds by using collective knowledge graph, leading to 33.3% efficiency improvement. The reason is that collective knowledge graph contains more information in entity, basic unit, maxterm, and rule layers. Once the entities "GPS" and "CPU" are extracted, their shielding statuses are determined without querying, which considerably reduces the interactive session rounds.

VI. Conclusion

An intelligent EMC diagnosis and management method was presented in this article. The mathematic sets were incorporated in the knowledge graph to enhance the searching efficiency and reduce the rounds of interactive sessions. At the same time, the EMC specified vocabularies were added in the thesaurus and trained, which improved the accuracy of entity extraction in the QA process. In addition, multiple entity extraction was achieved by decision trees. Although the test case is based on cell phones, the proposed method is applicable for all electronic products once their elements are added in the knowledge graph. The proposed method is practically useful for industrial product EMC design ahead of time.

The limitation of the method is the users must understand and collect all the related information before answering the queries by the system. In some cases, it is not so easy to understand the questions and know how to get the information. In the next step, the research will apply artificial intelligence on EMC feature extraction from the electronic design automation or computer-aided design files directly, then incorporate the information with the knowledge graph to realize the automation of EMC diagnosis and management.

References

[1] C. R. Paul, "A serious problem in using the Jarvis model for the prediction of crosstalk," *IEEE Trans. Electromagn. Compat.*, vol. 1, no. 3, pp. 46–55, Jul.–Sep. 2012.

[2] C.Y. Yang and V. Jandhyala, "A time-domain surface integral technique for mixed electromagnetic and circuit simulation," *IEEE Trans. Adv. Packag.*, vol. 28, no. 4, pp. 745–753, Nov. 2005.

[3] Q. He and D. Jiao, "Fast electromagnetics-based Co-simulation of linear network and nonlinear circuits for the analysis of high-speed integrated circuits," *IEEE Trans. Microw. Theory Techniq.*, vol. 58, no. 12, pp. 3677–3687, Dec. 2010.

[4] M. A. Azpurua, M. Pous, and F. Silva, "Decomposition of electromagnetic interferences in the time-domain," *IEEE Trans. Electromagn. Compat.*, vol. 58, no. 2, pp. 385–392, Apr. 2016.

[5] H. N. Lin, H. C. Chen, C. W. Kuo, and Y. T. Chang, "Design and application of a mobile miniature current probe for analysing the cause of EMI noise in IC circuits," *IET Sci., Measure. Technol.*, vol. 11, no. 5, pp. 655–665, Aug. 2017.

[6] J. C. S. Newell and H. A. Simon, "Report on a general problem-solving program," in *Proc. Int. Conf. Inf. Process.*, 1959, pp. 256–264.

[7] B. Chakravarti and M. Chakravarti, "Intelligence management of EMC control," in *Proc. Int. Conf. Electromagn. Interference Compat.*, 1999, pp. 156–159.

[8] M. Surekha, A. B. Patki, G. Radha, A. V. Sudha, G. S. Sekhar, and P. Shanthi, "Frame based knowledge systems for EMC analysis," in *Proc. IEEE Int. Symp. Electromagn. Compat.*, 1990, pp. 170–174.

[9] V. Choo, A. Drozd, D. Dixon, and N. Schade, "Implementation of intelligent EMC analysis and design techniques," in *Proc. IEEE Symp. Electromagn. Compat.*, 1994, pp. 167–172.

[10] V. Devabhaktuni, C. F. Bunting, D. Green, D. Kvale, L. Mareddy, and V. Rajamani, "A new ANN-based modeling approach for rapid EMI/EMC analysis of PCB and shielding enclosures," *IEEE Trans. Electromagn. Compat.*, vol. 55, no. 2, pp. 385–394, Apr. 2013.

[11] G. McCormick, Z. A. Khan, V. Devabhaktuni, M. Alam, and A. Wood, "Estimating radiated emissions from printed circuit boards and cables inside EMC chambers," in *Proc. IEEE Int. Symp. Electromagn. Compat.*, 2010, pp. 36–39.

[12] M. J. Kuo and T. C. Lin, "Dynamical optimal training for behavioral modeling of nonlinear circuit elements based on radial basis function neural network," in *Proc. Asia-Pacific Symp. Electromagn. Compat. 19th Int. Zurich Symp. Electromagn. Compat.*, 2008, pp. 670–673.

[13] I. Chahine, M. Kadi, E. Gaboriaud, A. Louis and B. Mazari, "Using neural networks for predicting the integrated circuits susceptibility to conducted electromagnetic disturbances," in *Proc. 18th Int. Zurich Symp. Electromagn. Compat.*, 2007, pp. 13–16.

[14] M. Magerl, C. Stockreiter, O. Eisenberger, R. Minixhofer, and A. Baric, "Building interchangeable black-box models of integrated circuits for EMC simulations," in *Proc. 10th Int. Workshop Electromagn. Compat. Integrated Circuits*, 2015, pp. 258–263.

[15] V. Ceperic, G. Gielen, and A. Baric, "Black-box modelling of conducted electromagnetic immunity by support vector machines," in *Proc. Int. Symp. Electromagn. Compat.*, 2012, pp. 1–6.

[16] R. Medico *et al.*, "Machine learning based error detection in transient susceptibility tests," *IEEE Trans. Electromagn. Compat.*, vol. 61, no. 2, pp. 352–360, Apr. 2019.

[17] A. Massa, D. Marcantonio, X. D. Chen, M. K. Li, and M. Salucci, "DNNs as applied to electromagnetics, antennas, and propagation - a review," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, no. 11, pp. 2225–2229, 2019.

[18] M. Nickel, K. Murphy, V. Tresp, and E. Gabrilovich, "A review of relational machine learning for knowledge graphs," in *Proc. IEEE*, 2016, pp. 11–33.

[19] S. Biswas, P. Mitra, and K. S. Rao, "Relation prediction of Co-morbid diseases using knowledge graph completion," *IEEE/ACM Trans. Comput. Biol. Bioinformat.*, to be published.

[20] X. L. Wang, Y. Wang, C. C. Gao, K. H. Lin, and Y. D. Li, "Automatic diagnosis with efficient medical case searching based on evolving graphs," *IEEE Access*, vol. 6, pp. 53307–53318, 2018.

[21] S. B. Shan and B.Y Cao, "Follow a guide to solve urban problems: the creation and application of urban knowledge graph," *IET Softw.*, vol. 11, no. 3, pp. 126–134, Jun. 2017.

[22] J. W. Qian, X. Y. Li, C. H. Zhang, L. L. Chen, T. Jung, and J. Z. Han, "Social network De-anonymization and privacy inference with knowledge graph model," *IEEE Trans. Dependable Secure Comput.*, vol. 16, no. 4, pp. 679–692, Aug. 2019.

[23] D. Shi, F. F. Zhang, N. Wang, and W. Fang, "Intelligent electromagnetic compatibility management of cell phones by using knowledge graphs," *IEEE Trans. Ind. Electron.*, vol. 66, no. 12, pp. 9808–9816, Dec. 2019.

[24] J. Pennington, R. Socher, and C. D. Manning, "GloVe: Global vectors for word representation," in *Proc. 2014 Conf. Empirical Methods Natural Language Processing (EMNLP)*, 2014, vol. 14, pp. 1532–1543.

[25] H. Palangi *et al.*, "Deep sentence embedding using long short-term memory networks: analysis and application to information retrieval," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 24. no. 4, pp. 694–707, Apr. 2016.

[26] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," in *Proc. 26th Annu. Conf. Neural Inf. Process. Syst.*, 2012, pp. 2951–2959.

[27] N. Reimers and I. Gurevych, "Optimal hyperparameters for deep LSTM-networks for sequence labeling tasks," *Comput. Lang.*, pp. 1–34, 2017. [Online]. Available: https://arxiv.org/abs/1707.06799